Assignment No. 7 – Interface and Abstract Class in Java

1. Interface in Java

An **interface** is a blueprint of a class that contains only abstract methods and constants.
It is used to achieve **100% abstraction** and supports **multiple inheritance**.

Key Features

- Declared using interface keyword
- Methods are **public and abstract by default**
- Variables are **public, static, final**
- Cannot have constructors
- A class uses implements keyword to implement interface
- Supports multiple inheritance

Example Explanation (SmartDevice)

- Created an interface **SmartDevice** with methods:
  - turnOn()
  - turnOff()
- Two classes implemented the interface:
  - **Fan**
  - **Light**
- Both classes provided their own implementation of methods.
- In the main class:

- Objects of Fan and Light are created
- Methods are called to display output

Output

Fan is ON
Light is ON
Fan is OFF
Light is OFF

Advantages of Interface

- Achieves full abstraction
- Supports multiple inheritance
- Improves code flexibility
- Promotes loose coupling

2. Abstract Class in Java

An **abstract class** is a class that cannot be instantiated.
It can contain both **abstract methods** and **non-abstract methods**.
Used to achieve **partial abstraction**.

Key Features

- Declared using abstract keyword
- Can have constructors
- Can have instance variables
- Can contain both abstract and normal methods
- Supports inheritance using extends

Example Explanation (BankAccount)

- Created an abstract class **BankAccount**:
  - Variable: balance
  - Constructor to initialize balance
  - Abstract method: calculateInterest()
  - Normal method: deposit()
- Created subclass **SavingsAccount**:
  - Inherits BankAccount
  - Implements calculateInterest() (4% interest)
- In the main class:
  - Object of SavingsAccount is created
  - Deposit method is called
  - Interest is calculated

Output

Deposited: 500.0
Savings Interest: 60.0

Advantages of Abstract Class

- Allows partial abstraction
- Code reusability
- Provides common base for classes
- Can include implemented methods

3. Difference Between Interface and Abstract Class

Feature	Interface	Abstract Class
Abstraction	100%	Partial
Methods	Only abstract	Abstract + Normal
Variables	Public static final	Instance variables allowed
Constructor	Not allowed	Allowed
Inheritance	Multiple	Single
Keyword	implements	extends

4. Conclusion

- **Interface** is used when only method declarations are required.
- **Abstract class** is used when some common functionality is shared.
- Both are important concepts for achieving abstraction in Java.